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Physical Data

General physical constants

Constant	Symbol	Value		
Speed of light in vacuum	c	2.99792458	$\times$	10^8 m s^{-1}
Gravitational constant	G	6.673(10)	$\times$	$10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Elementary charge	e	1.602176462(63)	$\times$	10^{-19} C
Avogadro constant	N_A	6.02214199(47)	$\times$	10^{23} mol^{-1}
Atomic mass unit or the Dalton (u or Da) ¹	u u c ²	1.66053873(13) 931.494061(21)	$\times$	10^{-27} kg MeV
Electron rest mass	m_o $m_o c^2$	9.10938188(72) 5.48579868(93) 0.510998928(11)	$\times$ $\times$	10^{-31} kg 10^{-4} u MeV
Proton rest mass	m_p $m_p c^2$	1.67262158(13) 1.00727646688(13) 938.272046(21)	$\times$	10^{-27} kg u MeV
Neutron rest mass	m_n $m_n c^2$	1.67492716(13) 1.00866491578(55) 939.565379(21)	$\times$	10^{-27} kg u MeV
Faraday constant	F	9.64853415(39)	$\times$	10^4 C mol^{-1}
Fine-structure constant α $\alpha = e^2 / 4\pi\epsilon_0 \hbar c$	α $1/\alpha$	7.297352533(27) 137.03599976(50)	$\times$	10^{-3}
Rydberg constant $me^4 / (4\pi\epsilon_0)^2 4\pi\hbar^3 c$	R_∞	1.0973731568549(83)	$\times$	10^7 m^{-1}
Bohr radius $\hbar^2 4\pi\epsilon_0 / m_o e^2$	a_o	5.291772083(19)	$\times$	10^{-11} m
Compton wavelength of electron $h/m_o c$	λ_c $\lambda_c / 2\pi$	2.426310215(88) 3.861592642(28)	$\times$ $\times$	10^{-12} m 10^{-13} m
Classical radius of electron $4\pi\epsilon_0 e^2 / mc^2$	r_o	2.817940285(13)	$\times$	10^{-15} m
Thomson cross section $8\pi r_o^2/3$	σ_T	6.65245854(15)	$\times$	10^{-29} m^2
Gyromagnetic ratio of proton	γ $\gamma/2\pi$	2.67522212(11) 4.25774825(18)	$\times$ $\times$	$10^8 \text{ rad s}^{-1} \text{ T}^{-1}$ 10^7 Hz T^{-1}
Bohr magneton $e\hbar / 2m_o$	μ_B	9.27400899(37)	$\times$	$10^{-24} \text{ J T}^{-1}$
Nuclear magneton $e\hbar / 2m_p$	μ_N	5.05078317(20)	$\times$	$10^{-27} \text{ J T}^{-1}$
Proton magnetic moment	μ_p μ_p/μ_N	1.410606633(58) 2.792847337(29)	$\times$	$10^{-26} \text{ J T}^{-1}$

¹ Atomic mass unit (u) and the Dalton (Da) are equivalent units.

Constants	Symbol	Value		
Gas constant	R	8.314462	×	$\text{J K}^{-1} \text{mol}^{-1}$
Boltzmann constant	k	1.380649	×	10^{-23}J K^{-1}
Stefan-Boltzmann constant	σ	5.670374	×	$10^{-8} \text{W m}^{-2} \text{K}^{-4}$
Permeability of free space	μ_0	4π	×	10^{-7}H m^{-1}
Permittivity of free space	ϵ_0	8.854187817	×	10^{-12}F m^{-1}
Planck constant	h	6.62607015	×	10^{-34}J s
	$\hbar$	1.054571817	×	10^{-34}J s
	$\hbar c$	197.3269718		MeV fermi
Charge-to-mass ratio for the electron	e/m_e	1.758820174(71)	×	10^{11}C kg^{-1}
Proton/electron mass		1.8361526675(39)	×	10^3
Nuclear radius (mass number A)		$\sim 1.2A^{1/3}$	×	10^{-15}m
Molar volume at S.T.P.	V_0	2.2413996(39)	×	$10^{-2} \text{m}^3 \text{mol}^{-1}$
		22.413996(39)		Litres mole^{-1}
Loschmidt's number	N_A/V_0	2.6867775(47)	×	10^{25}m^{-3}

1998 CODATA values, taken from *Rev Mod Phys*, 2000, **72**, 351-495

Conversion factors

Rydberg	1 Ry	$\simeq$	13.61 eV	atmosphere	1 atm	$\simeq$	$1.013 \times 10^5 \text{Pa}$
Atomic mass	1 u	$\simeq$	931.5 MeV/c ²	bar	1 bar	=	10^5Pa
unit (u)		$\simeq$	$1.6605 \times 10^{-27} \text{kg}$				
		$\simeq$	$1.6605 \times 10^{-27} \text{kg}$	torr	1 torr	=	1 mm Hg
Dalton	1 Da	$\simeq$	$1.6605 \times 10^{-27} \text{kg}$			$\simeq$	133.3 Pa
radian	1 rad	$\simeq$	57.296°	erg	1 erg	=	10^{-7}J
Ångström	1 Å	=	10^{-10}m	Gauss	1 G	=	10^{-4}T
Fermi	1 fm	=	10^{-15}m	Curie	1 Ci	=	37 ns ⁻¹
X unit	1 Xu	$\simeq$	$1.002 \times 10^{-13} \text{m}$	solar flux unit	1 sfu	=	$10^{-22} \text{Wm}^{-2}\text{Hz}^{-1}$
electronvolt	1 eV	$\simeq$	$1.602 \times 10^{-19} \text{J}$				
Jansky	1 Jy	=	$10^{-26} \text{Wm}^{-2}\text{Hz}^{-1}$				
barn	1 b	=	10^{-28}m^2				

Hartree atomic units

$$e = m = \hbar = 4\pi\epsilon_0 = 1$$

$$1 \text{ au (energy)} = 1 \text{ Hartree} = 2 \text{ Ry} = 27.21 \text{ eV}$$

$$1 \text{ au (length)} = a_0 = 5.292 \times 10^{-11} \text{ m}$$

The Greek Alphabet

A	α	Alpha	I	ι	Iota	P	ρ	Rho
B	β	Beta	K	κ	Kappa	Σ	σ	Sigma
Γ	γ	Gamma	Λ	λ	Lambda	T	τ	Tau
Δ	δ	Delta	M	μ	Mu	Y	υ	Upsilon
E	ε	Epsilon	N	ν	Nu	Φ	ϕ	Phi
Z	ζ	Zeta	Ξ	ξ	Xi	X	χ	Chi
H	η	Eta	O	$\omicron$	Omicron	Ψ	ψ	Psi
Θ	θ	Theta	Π	π	Pi	Ω	ω	Omega

Prefixes

<i>Prefix</i>	<i>Symbol</i>	<i>Factor</i>
yotta	Y	10^{24}
zetta	Z	10^{21}
exa	E	10^{18}
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
hecto	h	10^2
deca	da	10^1

<i>Prefix</i>	<i>Symbol</i>	<i>Factor</i>
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}
zepto	z	10^{-21}
yocto	y	10^{-24}

The Solar System

Body	Equatorial radius R/m	Mass M/kg	Density $\rho/kg\ m^{-3}$	Distance from Sun d/m	Surface Gravity $g/m\ s^{-2}$	No of Satellites N	Sidereal period T_s	Rotational period T_r
Mercury	2.44×10^6	3.301×10^{23}	5420	5.791×10^{10}	3.70	0	87.97 d	58.7 d
Venus	6.052×10^6	4.869×10^{24}	5250	1.082×10^{11}	8.85	0	224.7 d	243 d
Earth	6.378×10^6	5.978×10^{24}	5510	1.496×10^{11}	9.81	1	365.3 d	23.93 h
Mars	3.397×10^6	6.420×10^{23}	3960	2.279×10^{11}	3.72	2	687 d	24.6 h
Jupiter	7.14×10^7	1.899×10^{27}	1330	7.783×10^{11}	26.39	16	11.86 y	9.9 h
Saturn	6.00×10^7	5.685×10^{26}	680	1.427×10^{12}	11.67	18	29.46 y	10.2 h
Uranus	2.6×10^7	8.686×10^{25}	1600	2.869×10^{12}	9.12	17	84.02 y	10.7 h
Neptune	2.42×10^7	1.025×10^{26}	1650	4.498×10^{12}	13.8	8	164.8 y	15.8 h
Pluto	1.22×10^6	1.0×10^{22}	1400	5.900×10^{12}	0.49	1	248 y	6.3 d
Moon	1.74×10^6	7.348×10^{22}	3340	3.844×10^8 *	1.62	0	27.32 d	27.32 d
Sun	6.960×10^8	1.989×10^{30}						25 d (equator) 31 d (polar)

*From Earth

Solar Luminosity $L_{\text{Sun}} = 3.862 \times 10^{26}\ \text{W}$
Solar Effective Temperature $T_{\text{Sun}} = 5800\ \text{K}$

Astronomical Units

1 AU = $1.496 \times 10^{11}\ \text{m}$
1 parsec = $3.0857 \times 10^{16}\ \text{m}$
1 year = $3.156 \times 10^7\ \text{s}$

**Periodic Table:
with the Outer Electron Configurations of Neutral Atoms in their Ground States**

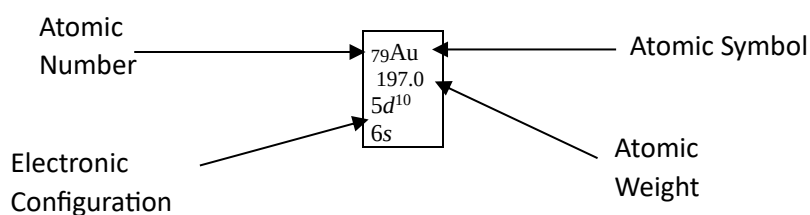
¹ H 1.00 1s																	² He 4.00 1s ²
³ Li 6.94 2s	⁴ Be 9.01 2s ²											⁵ B 10.81 2s ² 2p	⁶ C 12.01 2s ² 2p ²	⁷ N 14.01 2s ² 2p ³	⁸ O 16.00 2s ² 2p ⁴	⁹ F 19.00 2s ² 2p ⁵	¹⁰ Ne 20.18 2s ² 2p ⁶
¹¹ Na 22.99 3s	¹² Mg 24.21 3s ²											¹³ Al 26.98 3s ² 3p	¹⁴ Si 28.09 3s ² 3p ²	¹⁵ P 30.97 3s ² 3p ³	¹⁶ S 32.07 3s ² 3p ⁴	¹⁷ Cl 35.45 3s ² 3p ⁵	¹⁸ Ar 39.95 3s ² 3p ⁶
¹⁹ K 39.10 4s	²⁰ Ca 40.08 4s ²	²¹ Sc 44.96 3d 4s ²	²² Ti 47.88 3d ² 4s ²	²³ V 50.94 3d ³ 4s ²	²⁴ Cr 52.00 3d ⁵ 4s	²⁵ Mn 54.94 3d ⁵ 4s ²	²⁶ Fe 55.85 3d ⁶ 4s ²	²⁷ Co 58.93 3d ⁷ 4s ²	²⁸ Ni 58.69 3d ⁸ 4s ²	²⁹ Cu 63.55 3d ¹⁰ 4s	³⁰ Zn 65.39 3d ¹⁰ 4s ²	³¹ Ga 69.72 4s ² 4p	³² Ge 72.61 4s ² 4p ²	³³ As 74.92 4s ² 4p ³	³⁴ Se 78.96 4s ² 4p ⁴	³⁵ Br 79.90 4s ² 4p ⁵	³⁶ Kr 83.80 4s ² 4p ⁶
³⁷ Rb 85.47 5s	³⁸ Sr 87.62 5s ²	³⁹ Y 88.91 4d 5s ²	⁴⁰ Zr 91.22 4d ² 5s ²	⁴¹ Nb 92.91 4d ⁴ 5s	⁴² Mo 95.94 4d ⁵ 5s	⁴³ Tc 97.9 4d ⁶ 5s	⁴⁴ Ru 101.1 4d ⁷ 5s	⁴⁵ Rh 102.9 4d ⁸ 5s	⁴⁶ Pd 106.4 4d ¹⁰ -	⁴⁷ Ag 107.9 4d ¹⁰ 5s	⁴⁸ Cd 112.4 4d ¹⁰ 5s ²	⁴⁹ In 114.8 5s ² 5p	⁵⁰ Sn 118.7 5s ² 5p ²	⁵¹ Sb 121.8 5s ² 5p ³	⁵² Te 127.6 5s ² 5p ⁴	⁵³ I 126.9 5s ² 5p ⁵	⁵⁴ Xe 131.6 5s ² 5p ⁶
⁵⁵ Cs 132.9 6s	⁵⁶ Ba 137.3 6s ²	⁵⁷ La 138.9 5d 6s ²	⁷² Hf 178.5 4f ¹⁴ 5d ² 6s ²	⁷³ Ta 180.9 5d ³ 6s ²	⁷⁴ W 183.8 5d ⁴ 6s ²	⁷⁵ Re 186.2 5d ⁵ 6s ²	⁷⁶ Os 190.2 5d ⁶ 6s ²	⁷⁷ Ir 192.2 5d ⁹ -	⁷⁸ Pt 195.1 5d ⁹ 6s	⁷⁹ Au 197.0 5d ¹⁰ 6s	⁸⁰ Hg 200.6 5d ¹⁰ 6s ²	⁸¹ Tl 204.4 6s ² 6p	⁸² Pb 207.2 6s ² 6p ²	⁸³ Bi 209.0 6s ² 6p ³	⁸⁴ Po 209 6s ² 6p ⁴	⁸⁵ At 210 6s ² 6p ⁵	⁸⁶ Rn 222 6s ² 6p ⁶
⁸⁷ Fr 223 7s	⁸⁸ Ra 226.0 7s ²	⁸⁹ Ac 227.0 6d 7s ²	¹⁰⁴ Rf 267 /	¹⁰⁵ Db 268	¹⁰⁶ Sg 271	¹⁰⁷ Bh 272	¹⁰⁸ Hs 270	¹⁰⁹ Mt 276	¹¹⁰ Un 281	¹¹¹ Uu 272	¹¹² Uu 285						

lanthanides

actinides

⁵⁸ Ce 140.1 4f ² 6s ²	⁵⁹ Pr 140.9 4f ³ 6s ²	⁶⁰ Nd 144.2 4f ⁴ 6s ²	⁶¹ Pm 145 4f ⁵ 6s ²	⁶² Sm 150.4 4f ⁶ 6s ²	⁶³ Eu 152.0 4f ⁷ 6s ²	⁶⁴ Gd 157.3 4f ⁷ 5d 6s ²	⁶⁵ Tb 158.9 4f ⁸ 5d 6s ²	⁶⁶ Dy 162.5 4f ¹⁰ 6s ²	⁶⁷ Ho 164.9 4f ¹¹ 6s ²	⁶⁸ Er 167.3 4f ¹² 6s ²	⁶⁹ Tm 168.9 4f ¹³ 6s ²	⁷⁰ Yb 173.0 4f ¹⁴ 6s ²	⁷¹ Lu 175.0 4f ¹⁴ 5d 6s ²
⁹⁰ Th 232.0 - 6d ² 7s ²	⁹¹ Pa 231.0 5f ² 6d 7s ²	⁹² U 238.0 5f ³ 6d 7s ²	⁹³ Np 237.0 5f ⁴ 7s ²	⁹⁴ Pu 244 5f ⁶ 7s ²	⁹⁵ Am 243 5f ⁷ 7s ²	⁹⁶ Cm 247 5f ⁷ 6d 7s ²	⁹⁷ Bk 247	⁹⁸ Cf 251	⁹⁹ Es 252	¹⁰⁰ Fm 257	¹⁰¹ Md 258	¹⁰² No 259	¹⁰³ Lr 260

Legend: The notation used to describe the electronic configuration of atoms and ions is discussed in all textbooks of introductory atomic physics. The letters s, p, d, ... signify electrons having orbital angular momentum 0, 1, 2, ... in units $\hbar$; the number to the left of the letter denotes the principal quantum number of one orbit, and the superscript to the right denotes the number of electrons in the orbit.



Mathematical Formulae

Sum of a Geometric Series

$$\begin{aligned} S_n &= a(1 + r + r^2 + \dots + r^n) = a \frac{1 - r^{n+1}}{1 - r} \\ &= a(n + 1) \quad (r = 1) \\ &\rightarrow a \frac{1}{1 - r} \quad n \rightarrow \infty, |r| < 1 \end{aligned}$$

Integration by Parts

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

Stirling's Formula

$$\ln(N!) = N \ln(N) - N \quad (N, \text{ large integer})$$

Binomial Coefficient

$$\binom{n}{r} = {}^n C_r = \frac{n!}{r! (n - r)!} = \frac{n(n - 1) \dots (n - r + 1)}{r!}$$

Binomial Theorem

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + b^n$$

$$(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha - 1)}{2!} x^2 + \dots \quad (|x| < 1)$$

Definite integrals

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}} \quad (a > 0, n \text{ positive integer}) \quad \int_0^{\infty} \frac{\sin(|p|x)}{x} dx = \frac{\pi}{2} \quad (p \neq 0)$$

$$\int_0^{\infty} e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}} \quad (a > 0) \quad \int_0^{\infty} \frac{\sin^2(px)}{x^2} dx = \frac{\pi p}{2}$$

$$\int_0^{\infty} x^2 e^{-ax^2} dx = \frac{1}{4} \sqrt{\frac{\pi}{a^3}} \quad \int_0^{\pi} \sin(mx) \sin(nx) dx = \begin{cases} 0, m \neq n \\ \frac{\pi}{2}, m = n \end{cases}$$

$$\int_0^{\infty} x^4 e^{-ax^2} dx = \frac{3}{8} \sqrt{\frac{\pi}{a^5}} \quad \int_0^{\pi} \cos(mx) \cos(nx) dx = \begin{cases} 0, m \neq n \\ \frac{\pi}{2}, m = n \end{cases}$$

$$\int_0^{\infty} x e^{-ax^2} dx = \frac{1}{2a} \quad (a > 0) \quad \int_0^{\pi} \sin(mx) \cos(nx) dx = \begin{cases} 0, (m+n) \text{ even} \\ \frac{2m}{m^2 - n^2}, (m+n) \text{ odd} \end{cases}$$

$$\int_0^{\infty} x^3 e^{-ax^2} dx = \frac{1}{2a^2} \quad (a > 0) \quad \iint \frac{1}{r_{12}} e^{-2a(r_1+r_2)} d^3\mathbf{r}_1 d^3\mathbf{r}_2 = \frac{5\pi^2}{8a^5}$$

$$\int_0^{\infty} x^5 e^{-ax^2} dx = \frac{1}{a^3} \quad (a > 0)$$

$$\int_0^{\infty} \frac{x}{e^x - 1} dx = \frac{\pi^2}{6}$$

$$\int_0^{\infty} \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}$$

$$\int_0^{\infty} \frac{x^4 e^x}{(e^x - 1)^2} dx = \frac{4\pi^4}{15}$$

Indefinite integrals

Function of x	Integral with respect to x
x^n $\frac{1}{x} = x^{-1}$ e^x a^x	$\frac{x^{n+1}}{n+1} \quad (n \neq -1)$ $\log_e(x)$ e^x $\frac{a^x}{\log_e(a)}$
$\cos x$ $\sin x$ $\sec^2 x = \frac{1}{\cos^2 x}$ $\sec x \tan x = \frac{\sin x}{\cos^2 x}$ $\operatorname{cosec} x \cot x = \frac{\cos x}{\sin^2 x}$ $\operatorname{cosec}^2 x = 1/\sin^2 x$	$\sin x$ $-\cos x$ $\tan x$ $\sec x$ $-\operatorname{cosec} x$ $-\cot x$
$\frac{1}{\sqrt{1-x^2}} \quad (x^2 < 1)$ $\frac{1}{x\sqrt{x^2-1}} \quad (x^2 > 1)$ $\frac{1}{1+x^2}$	$\sin^{-1} x \text{ or } -\cos^{-1} x$ $\sec^{-1} x \text{ or } -\operatorname{cosec}^{-1} x$ $\tan^{-1} x \text{ or } -\cot^{-1} x$

Function of x	Integral with respect to x
$(ax+b)^n$ $\frac{1}{ax+b} = (ax+b)^{-1}$ e^{ax+b} a^{bx+c}	$\frac{(ax+b)^{n+1}}{a(n+1)} \quad (n \neq -1)$ $\frac{1}{a} \log_e(ax+b)$ $\frac{1}{a} e^{ax+b}$ $\frac{1}{b \log_e a} a^{bx+c}$
$\cos(ax+b)$ $\sin(ax+b)$ $\sec^2(ax+b) = \frac{1}{\cos^2(ax+b)}$ $\sec(ax+b) \tan(ax+b)$ $\operatorname{cosec}(ax+b) \cot(ax+b)$ $\operatorname{cosec}^2(ax+b)$	$\frac{1}{a} \sin(ax+b)$ $-\frac{1}{a} \cos(ax+b)$ $\frac{1}{a} \tan(ax+b)$ $\frac{1}{a} \sec(ax+b)$ $-\frac{1}{a} \operatorname{cosec}(ax+b)$ $-\frac{1}{a} \cot(ax+b)$
$\frac{1}{\sqrt{a^2-x^2}} \quad (x^2 < a^2)$ $\frac{1}{x\sqrt{x^2-a^2}} \quad (x^2 > a^2)$ $\frac{1}{a^2+x^2}$	$\sin^{-1} \frac{x}{a} \text{ or } -\cos^{-1} \frac{x}{a}$ $\frac{1}{a} \sec^{-1} \frac{x}{a} \text{ or } -\frac{1}{a} \operatorname{cosec}^{-1} \frac{x}{a}$ $\frac{1}{a} \tan^{-1} \frac{x}{a} \text{ or } -\frac{1}{a} \cot^{-1} \frac{x}{a}$

Vector relationships

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

$$\nabla(\theta\varphi) = (\nabla\theta)\varphi + \theta(\nabla\varphi)$$

$$\nabla \cdot (\varphi \mathbf{A}) = (\nabla\varphi) \cdot \mathbf{A} + \varphi(\nabla \cdot \mathbf{A})$$

$$\nabla \times (\varphi \mathbf{A}) = (\nabla\varphi) \times \mathbf{A} + \varphi(\nabla \times \mathbf{A})$$

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} + (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{B} \times (\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla \times \mathbf{B})$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$$

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

$$\nabla \times \nabla\varphi = 0$$

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

$$\int_V \nabla\varphi \, dV = \oint_S \varphi \, d\mathbf{S}$$

$$\int_V \nabla \cdot \mathbf{A} \, dV = \oint_S \mathbf{A} \cdot d\mathbf{S}$$

$$\int_V (\nabla \times \mathbf{A}) \, dV = \oint_S d\mathbf{S} \times \mathbf{A}$$

$$\int_S d\mathbf{S} \times \nabla\varphi = \oint_C \varphi \, d\boldsymbol{\ell}$$

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$$

$$\int_S (d\mathbf{S} \times \nabla) \times \mathbf{A} = \oint_C d\boldsymbol{\ell} \times \mathbf{A}$$

$$\int_V (\nabla\varphi \cdot \nabla\theta + \varphi \nabla^2\theta) \, dV = \oint_S \varphi \nabla\theta \cdot d\mathbf{S}$$

$$\int_V (\varphi \nabla^2\psi - \psi \nabla^2\varphi) \, dV = \oint_S (\varphi \nabla\psi - \psi \nabla\varphi) \cdot d\mathbf{S}$$

Grad, Div, Curl and Laplacian for various coordinate systems

Cartesian coordinates

$$\text{Gradient} \quad \nabla V = \hat{x} \frac{\partial V}{\partial x} + \hat{y} \frac{\partial V}{\partial y} + \hat{z} \frac{\partial V}{\partial z}$$

$$\text{Divergence} \quad \nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\text{Curl} \quad \nabla \times \mathbf{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\text{Laplacian} \quad \nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\nabla^2 \mathbf{A} = \hat{x} \nabla^2 A_x + \hat{y} \nabla^2 A_y + \hat{z} \nabla^2 A_z$$

Cylindrical polar coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

$$\text{Gradient} \quad \nabla V = \hat{r} \frac{\partial V}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial V}{\partial \theta} + \hat{z} \frac{\partial V}{\partial z}$$

$$\text{Divergence} \quad \nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial}{\partial \theta} A_\theta + \frac{\partial}{\partial z} A_z$$

$$\text{Curl} \quad \nabla \times \mathbf{A} = \frac{1}{r} \begin{vmatrix} \hat{r} & r \hat{\theta} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ A_r & r A_\theta & A_z \end{vmatrix}$$

$$\text{Laplacian} \quad \nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical polar coordinates

$$x = r \sin\theta \cos\varphi \quad y = r \sin\theta \sin\varphi \quad z = r \cos\theta$$

$$\text{Gradient} \quad \nabla V = \hat{\mathbf{r}} \frac{\partial V}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial V}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{r \sin\theta} \frac{\partial V}{\partial \varphi}$$

$$\text{Divergence} \quad \nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\sin\theta A_\theta) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \varphi} A_\varphi$$

$$\text{Curl} \quad \nabla \times \mathbf{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \hat{\mathbf{r}} & r\hat{\boldsymbol{\theta}} & r\sin\theta\hat{\boldsymbol{\phi}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ A_r & rA_\theta & r\sin\theta A_\varphi \end{vmatrix}$$

$$\text{Laplacian} \quad \nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 V}{\partial \varphi^2}$$

Sum, difference and product of trigonometric functions

$$\sin(A) + \sin(B) = 2 \sin\left(\frac{1}{2}(A+B)\right) \cos\left(\frac{1}{2}(A-B)\right)$$

$$\sin(A) - \sin(B) = 2 \cos\left(\frac{1}{2}(A+B)\right) \sin\left(\frac{1}{2}(A-B)\right)$$

$$\sin(A+B) = \sin(A) \cos(B) + \cos(A) \sin(B)$$

$$\sin(A-B) = \sin(A) \cos(B) - \cos(A) \sin(B)$$

$$\cos(A) + \cos(B) = 2 \cos\left(\frac{1}{2}(A+B)\right) \cos\left(\frac{1}{2}(A-B)\right)$$

$$\cos(A) - \cos(B) = 2 \sin\left(\frac{1}{2}(A+B)\right) \sin\left(\frac{1}{2}(B-A)\right)$$

$$\cos(A+B) = \cos(A) \cos(B) - \sin(A) \sin(B)$$

$$\cos(A-B) = \cos(A) \cos(B) + \sin(A) \sin(B)$$

$$\sin(A) \sin(B) = \frac{1}{2} \{ \cos(A-B) - \cos(A+B) \}$$

$$\tan(A+B) = \frac{\tan(A) + \tan(B)}{1 - \tan(A) \tan(B)}$$

$$\tan(A-B) = \frac{\tan(A) - \tan(B)}{1 + \tan(A) \tan(B)}$$

$$\cos(A) \cos(B) = \frac{1}{2} \{ \cos(A - B) + \cos(A + B) \}$$

$$\sin(A) \cos(B) = \frac{1}{2} \{ \sin(A - B) + \sin(A + B) \}$$

$$\sin^2(A) = \frac{1}{2} (1 - \cos(2A))$$

$$\sin(2A) = 2 \cos(A) \sin(A)$$

$$\cos(2A) = \cos^2(A) - \sin^2(A)$$

$$\cos^2(A) = \frac{1}{2} (1 + \cos(2A))$$

Standard Differential Equations

Bessel's Equation

$$x^2 y'' + xy' + (x^2 - n^2)y = 0$$

Legendre's Equation

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$$

Associated Legendre Equation

$$(1 - x^2)y'' - 2xy' + \left[n(n + 1) - \frac{m^2}{1 - x^2} \right] y = 0$$

Hermite's Equation

$$y'' - xy' + ny = 0$$

Laguerre's Equation

$$xy'' + (a + 1 - x)y' + ny = 0$$

Recurrence relations

Bessel Functions

$$J_{m-1}(x) + J_{m+1}(x) = \frac{2m}{x} J_m(x)$$

Legendre Functions

$$(n + 1)P_{n+1}(x) - (2n + 1)xP_n(x) + nP_{n-1}(x) = 0$$

Associated Legendre Functions:

$$(n + 1 - m)P_{n+1}^m(x) - (2n + 1)xP_n^m(x) + (n + m)P_{n-1}^m(x) = 0$$

$$P_n^{m+2}(x) - \frac{2(m + 1)x}{(1 - x^2)^{\frac{1}{2}}} P_n^{m+1}(x) + (n - m)(n + m + 1)P_n^m(x) = 0$$

Hermite Functions

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$$

Legendre polynomials

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1)$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x)$$

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)$$

Associated Legendre polynomials

$$P_1^1(x) = (1 - x^2)^{1/2}$$

$$P_2^1(x) = 3x(1 - x^2)^{1/2}$$

$$P_2^2(x) = 3(1 - x^2)$$

$$P_3^1(x) = \frac{3}{2}(5x^2 - 1)(1 - x^2)^{1/2}$$

$$P_3^2(x) = 15x(1 - x^2)$$

$$P_3^3(x) = 15(1 - x^2)^{3/2}$$

Hermite polynomials

$$H_0(x) = 1$$

$$H_1(x) = 2x$$

$$H_2(x) = 4x^2 - 2$$

$$H_3(x) = 8x^3 - 12x$$

$$H_4(x) = 16x^4 - 48x^2 + 12$$

Fourier transforms

Function²

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

$$a_1 f_1(t) + a_2 f_2(t)$$

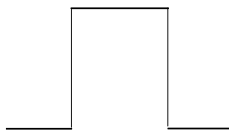
$$f(-t)$$

$$f(t - t_0)$$

$$f(t) e^{i\omega_0 t}$$

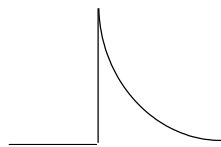
$$f'(t)$$

$$f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(x) f_2(t - x) dx$$



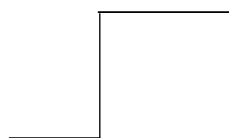
$$\begin{cases} \frac{1}{T} & \left(-\frac{T}{2} < t < \frac{T}{2}\right) \\ 0 & \text{for other } t \end{cases}$$

$$\frac{\sin(\omega T/2)}{(\omega T/2)}$$



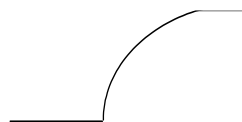
$$\begin{cases} e^{-\alpha t} & (t > 0, \alpha > 0) \\ 0 & \text{for all other } t \end{cases}$$

$$\frac{1}{\alpha + i\omega}$$



$$\begin{cases} 1 & (t > 0) \\ 0 & (t < 0) \end{cases}$$

$$\frac{1}{i\omega} + \pi\delta(\omega)$$



$$\begin{cases} 1 - e^{-\alpha t} & (t > 0) \\ 0 & (t < 0) \end{cases}$$

$$\frac{1}{i\omega} - \frac{1}{\alpha + i\omega} = \frac{\alpha}{i\omega(\alpha + i\omega)}$$

$$e^{-\alpha|t|}$$

$$\alpha > 0$$

$$\frac{2\alpha}{\alpha^2 + \omega^2}$$

$$e^{-\alpha t^2}$$

$$\alpha > 0$$

$$\sqrt{\frac{\pi}{\alpha}} e^{-\frac{\omega^2}{4\alpha}}$$

$$\begin{cases} e^{-\alpha t} \sin(bt) & \\ 0 & \end{cases}$$

$$\begin{cases} \alpha > 0, t > 0 \\ t \leq 0 \end{cases}$$

$$\frac{b}{(i\omega + \alpha)^2 + b^2}$$

Fourier Transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$a_1 F(\omega) + a_2 F(\omega)$$

$$F(-\omega)$$

$$F(\omega) e^{-i\omega t_0}$$

$$F(\omega - \omega_0)$$

$$i\omega F(\omega)$$

$$F_1(\omega) F_2(\omega)$$

² Where $i = \sqrt{-1}$ is the imaginary number identity.

Function³**Fourier Transform**

$$\begin{cases} e^{-\alpha t} \cos(bt) \\ 0 \end{cases}$$

$$\begin{cases} \alpha > 0, t > 0 \\ t \leq 0 \end{cases}$$

$$\frac{i\omega + \alpha}{(i\omega + \alpha)^2 + b^2}$$

$$\delta(t)$$

$$1$$

$$\delta(t - t_0)$$

$$e^{i\omega t_0}$$

Laplace Transforms

$$f(x)$$

$$\overline{f(s)} = \int_0^\infty e^{-sx} f(x) dx$$

$$f = \mathcal{L}^{-1} \bar{f}$$

$$\bar{f} = \mathcal{L}f$$

$$1$$

$$\frac{1}{s}$$

$$x$$

$$\frac{1}{s^2}$$

$$x^n$$

$$\frac{n!}{s^{n+1}} \quad (n \text{ integer}, n \geq 0)$$

$$e^{-\alpha x}$$

$$\frac{1}{s + \alpha}$$

$$e^{\alpha x}$$

$$\frac{1}{s - \alpha}$$

$$\frac{1}{2}(e^{\alpha x} - e^{-\alpha x}) = \sinh(\alpha x)$$

$$\frac{\alpha}{s^2 - \alpha^2}$$

$$\frac{1}{2}(e^{\alpha x} + e^{-\alpha x}) = \cosh(\alpha x)$$

$$\frac{s}{s^2 - \alpha^2}$$

$$\frac{1}{2i}(e^{i\alpha x} - e^{-i\alpha x}) = \sin(\alpha x)$$

$$\frac{\alpha}{s^2 + \alpha^2}$$

$$\frac{1}{2}(e^{i\alpha x} + e^{-i\alpha x}) = \cos(\alpha x)$$

$$\frac{s}{s^2 + \alpha^2}$$

$$e^{\alpha x} f(x)$$

$$f(s - \alpha)$$

$$e^{\alpha x} x^n$$

$$\frac{n!}{(s - \alpha)^{n+1}} \quad (n \text{ integer}, n \geq 0)$$

$$e^{\alpha x} \sin(\beta x)$$

$$\frac{\beta}{(s - \alpha)^2 + \beta^2}$$

³ Where $i = \sqrt{-1}$ is the imaginary number identity.

$$e^{\alpha x} \cos(\beta x)$$

$$\frac{s - \beta}{(s - \alpha)^2 + \beta^2}$$

Convolution

$$\int_0^x f(\lambda)g(x - \lambda)d\lambda$$

$$\overline{f(s)} \cdot \overline{g(s)}$$

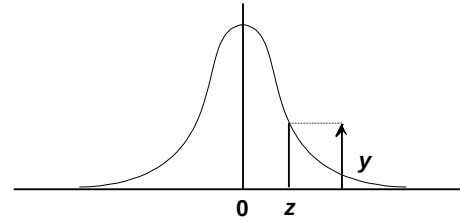
Derivatives

$$\frac{d^n y}{dx^n}$$

$$s^n \bar{y} - s^{n-1}y(0) - s^{n-2}y^1(0) \dots y^{n-1}(0)$$

Ordinates of a standard normal curve

$$y = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

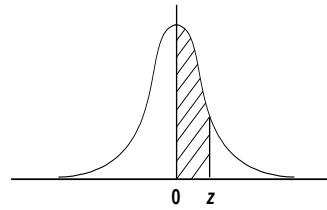


z	0	1	2	3	4	5	6	7	8	9
0.0	.3989	.3989	.3989	.3988	.3986	.3984	.3982	.3980	.3977	.3973
0.1	.3970	.3965	.3961	.3956	.3951	.3945	.3939	.3932	.3925	.3918
0.2	.3910	.3902	.3894	.3885	.3876	.3867	.3857	.3847	.3836	.3825
0.3	.3814	.3802	.3790	.3778	.3765	.3752	.3739	.3725	.3712	.3697
0.4	.3683	.3668	.3653	.3637	.3621	.3605	.3589	.3572	.3555	.3538
0.5	.3521	.3503	.3485	.3467	.3448	.3429	.3410	.3391	.3372	.3352
0.6	.3332	.3312	.3292	.3271	.3251	.3230	.3209	.3187	.3166	.3144
0.7	.3123	.3101	.3079	.3056	.3034	.3011	.2989	.2966	.2943	.2920
0.8	.2897	.2874	.2850	.2827	.2803	.2780	.2756	.2732	.2709	.2685
0.9	.2661	.2637	.2613	.2589	.2565	.2541	.2516	.2492	.2468	.2444
1.0	.2420	.2396	.2371	.2347	.2323	.2299	.2275	.2251	.2227	.2203
1.1	.2179	.2155	.2131	.2107	.2083	.2059	.2036	.2012	.1989	.1965
1.2	.1942	.1919	.1895	.1872	.1849	.1826	.1804	.1781	.1758	.1736
1.3	.1714	.1691	.1669	.1647	.1626	.1604	.1582	.1561	.1539	.1518
1.4	.1497	.1476	.1456	.1435	.1415	.1394	.1374	.1354	.1334	.1315
1.5	.1295	.1276	.1257	.1238	.1219	.1200	.1182	.1163	.1145	.1127
1.6	.1109	.1092	.1074	.1057	.1040	.1023	.1006	.0989	.0973	.0957
1.7	.0940	.0925	.0909	.0893	.0878	.0863	.0848	.0833	.0818	.0804
1.8	.0790	.0775	.0761	.0748	.0734	.0721	.0707	.0694	.0681	.0669
1.9	.0656	.0644	.0632	.0620	.0608	.0596	.0584	.0573	.0562	.0551
2.0	.0540	.0529	.0519	.0508	.0498	.0488	.0478	.0468	.0459	.0449
2.1	.0440	.0431	.0422	.0413	.0404	.0396	.0387	.0379	.0371	.0363
2.2	.0355	.0347	.0339	.0332	.0325	.0317	.0310	.0303	.0297	.0290
2.3	.0283	.0277	.0270	.0264	.0258	.0252	.0246	.0241	.0235	.0229
2.4	.0224	.0219	.0213	.0208	.0203	.0198	.0194	.0189	.0184	.0180
2.5	.0175	.0171	.0167	.0163	.0158	.0154	.0151	.0147	.0143	.0139
2.6	.0136	.0132	.0129	.0126	.0122	.0119	.0116	.0113	.0110	.0107
2.7	.0104	.0101	.0099	.0096	.0093	.0091	.0088	.0086	.0084	.0081
2.8	.0079	.0077	.0075	.0073	.0071	.0069	.0067	.0065	.0063	.0061
2.9	.0060	.0058	.0056	.0055	.0053	.0051	.0050	.0048	.0047	.0046
3.0	.0044	.0043	.0042	.0040	.0039	.0038	.0037	.0036	.0035	.0034
3.1	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026	.0025	.0025
3.2	.0024	.0023	.0022	.0022	.0021	.0020	.0020	.0019	.0018	.0018
3.3	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014	.0013	.0013
3.4	.0012	.0012	.0012	.0011	.0011	.0010	.0010	.0010	.0009	.0009
3.5	.0009	.0008	.0008	.0008	.0008	.0007	.0007	.0007	.0007	.0006
3.6	.0006	.0006	.0006	.0005	.0005	.0005	.0005	.0005	.0005	.0004
3.7	.0004	.0004	.0004	.0004	.0004	.0004	.0003	.0003	.0003	.0003
3.8	.0003	.0003	.0003	.0003	.0003	.0002	.0002	.0002	.0002	.0002
3.9	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0001	.0001

Columns 0-9 refer to the second decimal place in z.

Area under a standard normal curve from 0 to z

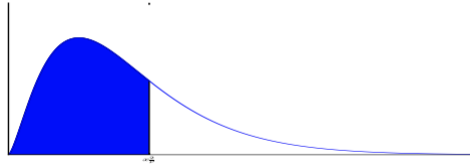
$$\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2} dz$$



z	0	1	2	3	4	5	6	7	8	9
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0754
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2258	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.2612	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2996	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4843	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998
3.6	.4998	.4998	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.7	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.8	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.9	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000

Columns 0-9 refer to the second decimal place in z.

Percentile values of χ^2_p



For the chi-squared distribution with x degrees of freedom (shaded area = p)

χ	$\chi_{\frac{2}{0.995}}$	$\chi_{\frac{2}{0.990}}$	$\chi_{\frac{2}{0.975}}$	$\chi_{\frac{2}{0.950}}$	$\chi_{\frac{2}{0.900}}$	$\chi_{\frac{2}{0.750}}$	$\chi_{\frac{2}{0.500}}$	$\chi_{\frac{2}{0.250}}$	$\chi_{\frac{2}{0.100}}$	$\chi_{\frac{2}{0.050}}$	$\chi_{\frac{2}{0.025}}$	$\chi_{\frac{2}{0.010}}$	$\chi_{\frac{2}{0.005}}$
1	3.03e-5	1.57e-4	9.82e-4	3.93e-3	0.01579	0.1015	0.4549	1.323	2.706	3.841	5.024	6.635	7.879
2	0.01002	0.0201	0.05064	0.1026	0.2107	0.5754	1.386	2.773	4.605	5.991	7.378	9.21	10.6
3	0.07171	0.1148	0.2158	0.3518	0.5844	1.213	2.366	4.108	6.251	7.815	9.348	11.34	12.84
4	0.207	0.2971	0.4844	0.7107	1.064	1.923	3.357	5.385	7.779	9.488	11.14	13.28	14.86
5	0.4118	0.5543	0.8312	1.145	1.61	2.675	4.351	6.626	9.236	11.07	12.83	15.09	16.75
6	0.6757	0.8721	1.237	1.635	2.204	3.455	5.348	7.841	10.64	12.59	14.45	16.81	18.55
7	0.9892	1.239	1.69	2.167	2.833	4.255	6.346	9.037	12.02	14.07	16.01	18.48	20.28
8	1.344	1.647	2.18	2.733	3.49	5.071	7.344	10.22	13.36	15.51	17.53	20.09	21.95
9	1.735	2.088	2.7	3.325	4.168	5.899	8.343	11.39	14.68	16.92	19.02	21.67	23.59
10	2.156	2.558	3.247	3.94	4.865	6.737	9.342	12.55	15.99	18.31	20.48	23.21	25.19
11	2.603	3.053	3.816	4.575	5.578	7.584	10.34	13.7	17.27	19.68	21.92	24.72	26.76
12	3.074	3.571	4.404	5.226	6.304	8.438	11.34	14.85	18.55	21.03	23.34	26.22	28.3
13	3.565	4.107	5.009	5.892	7.041	9.299	12.34	15.98	19.81	22.36	24.74	27.69	29.82
14	4.075	4.66	5.629	6.571	7.79	10.17	13.34	17.12	21.06	23.68	26.12	29.14	31.32
15	4.601	5.229	6.262	7.261	8.547	11.04	14.34	18.25	22.31	25	27.49	30.58	32.8
16	5.142	5.812	6.908	7.962	9.312	11.91	15.34	19.37	23.54	26.3	28.85	32	34.27
17	5.697	6.408	7.564	8.672	10.09	12.79	16.34	20.49	24.77	27.59	30.19	33.41	35.72
18	6.265	7.015	8.231	9.39	10.86	13.68	17.34	21.6	25.99	28.87	31.53	34.81	37.16
19	6.844	7.633	8.906	10.12	11.65	14.56	18.34	22.72	27.2	30.14	32.85	36.19	38.58
20	7.434	8.26	9.591	10.85	12.44	15.45	19.34	23.83	28.41	31.41	34.17	37.57	40
21	8.034	8.897	10.28	11.59	13.24	16.34	20.34	24.93	29.62	32.67	35.48	38.93	41.4
22	8.643	9.542	10.98	12.34	14.04	17.24	21.34	26.04	30.81	33.92	36.78	40.29	42.8
23	9.26	10.2	11.69	13.09	14.85	18.14	22.34	27.14	32.01	35.17	38.08	41.64	44.18
24	9.886	10.86	12.4	13.85	15.66	19.04	23.34	28.24	33.2	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	19.94	24.34	29.34	34.38	37.65	40.65	44.31	46.93
26	11.16	12.2	13.84	15.38	17.29	20.84	25.34	30.43	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	21.75	26.34	31.53	36.74	40.11	43.19	46.96	49.64
28	12.46	13.56	15.31	16.93	18.94	22.66	27.34	32.62	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	23.57	28.34	33.71	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.6	24.48	29.34	34.8	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	33.66	39.34	45.62	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	42.94	49.33	56.33	63.17	67.5	71.42	76.15	79.49
60	35.53	37.49	40.48	43.19	46.46	52.29	59.33	66.98	74.4	79.08	83.3	88.38	91.95
70	43.27	45.44	48.76	51.74	55.33	61.7	69.33	77.58	85.53	90.53	95.02	100.4	104.2
80	51.17	53.54	57.15	60.39	64.28	71.14	79.33	88.13	96.58	101.9	106.6	112.3	116.3
90	59.2	61.75	65.65	69.13	73.29	80.62	89.33	98.65	107.6	113.1	118.1	124.1	128.3
100	67.33	70.06	74.22	77.93	82.36	90.13	99.33	109.1	118.5	124.3	129.6	135.8	140.2